

CODECHECK certificate 2025-023

github.com/codecheckers/certificate-2025-023/



Table 1: CODECHECK summary

Item	Value
Title	<i>A calibrated optogenetic toolbox of stable zebrafish opsin lines</i>
Authors	Paride Antinucci  , Adna Dumitrescu  , Charlotte Deleuze, Holly J Morley  , Kristie Leung, Tom Hagley, Fumi Kubo, Herwig Baier  , Isaac H Bianco  , Claire Wyart 
Reference	doi.org/10.7554/eLife.54937
Repository	github.com/codecheckers/certificate-2025-023
Codechecker	Linus Dexter Hackel 
Date of check	2026-01-15
Summary	Figures 4, 5, 8 and 9 could be reproduced partially. There was no documentation how Figures 1, 2, 3, 6 and 10 were produced or with which scripts, so they couldn't be reproduced.

Table 2: Summary of output files generated

File	Comment	Size (b)
Gapfree_AP_stim.csv	The .csv file containing the gap free AP stimulation.	1870
figure4_1.pdf	Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)	16548
figure4_2.pdf	Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chrimson)	82263
figure4_3.pdf	Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)	63693
figure5_1.pdf	Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)	78091
figure5_1.pdf	Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chrimson)	78091
figure8_1.pdf	Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)	235116

File	Comment	Size (b)
figure8_2.pdf	Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)	213717
figure9_1.pdf	Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)	245474
figure9_2.pdf	Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)	248817
figure9_3.pdf	Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)	179181
figure9_4.pdf	Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)	94949
figure9_5.pdf	Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)	121586

Summary

Figures 4, 5, 8 and 9 could be reproduced partially. There was no documentation how Figures 1, 2, 3, 6 and 10 were produced or with which scripts, so they couldn't be reproduced.

CODECHECKER notes

Environment

Setting up the environment took a bit of time, as older versions of Python and older Dependencies needed to be properly installed, but it is all very well documented in the README file what dependencies need to be installed and with which version.

Since I didn't want a new window for the plot results of Matplotlib, I decided to put the following sequence at the end of each script, so the created figure is automatically saved to the *outputs/figures/* directory.

```
figure_number = 9

figure_id = int(input('Please enter the Figure ID.\n\n'))

plt.savefig(f"figures/figure{figure_number}_{figure_id}.pdf")
```

Script Errors

In the script *Excitatory_Opsin_Current_Clamp.py* the filepath for the trace contained a typo in line 452. It was: *'Analysis_output/Single_trace_data/CC_excitatory/2019_03_19_0055.csv'* but it should have been *'Analysis_output/Single_Trace_data/CC_excitatory/2019_03_19_0055.csv'*. This is just a small error, but it still took me some minutes in figuring out, where the *FileNotFoundError* could be coming from, as the file appeared to be there.

Similar to the first error, in the script *Inhibitory_Opsin_Current_Clamp.py* the filepath for the CC inhibitory opsin master sheet had the wrong filepath and needed to be changed in the lines 400, 417, 423. It was: *'/Users/adna.dumitrescu/Documents/Wyart_Postdoc/Data/OPsin_testing_project/Opsin_Ephys_Analysis/CC_analysis/CC_opsin_inhibitory_master.csv'* but it should have been: *'Analysis_output/CC_opsin_inhibitory_master.csv'*. Different to the first error though, the file didn't exist entirely, so it had to be newly created with the same *csv-Header* as the file

'Analysis_output/VC_inhibitory_opsin_master.csv'. After these two fixes, the script worked perfectly.

In the script *Inhibitory_Opsin_CC_Long_AP_Inhibit.py* the line 424 had to be changed from an array of the length 7 to an array of the length 2, as this was what was given in the documentation and also what the code afterwards expected. The code was therefore changed from:

```
LED_max_V_user = [input('pulse_1: \n'), input('pulse2: \n'), input('pulse3: \n'),  
input('pulse4: \n'), input('pulse5: \n'), input('pulse6: \n'), input('pulse7:  
\n')]
```

to

```
LED_max_V_user = [input('pulse_1: \n'), input('pulse2: \n')]
```

Recommendations to the authors

TODO

Citing this document

Linus Dexter Hackel (2026). CODECHECK Certificate 2025-023. Zenodo. github.com/codecheckers/certificate-2025-023/

About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely available by following the links in this document, the reader can then study for themselves the code and data.

About this document

This document was created using a [jupyter notebook](#) and converted into Markdown via [nbconvert](#) and [pandoc](#). Afterwards it was converted into [Typst](#) using [cmarker](#) and then into PDF using Typst.

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Manifest files

CSV files

Analysis_output/Gapfree_AP_stim.csv

Author comment: *The .csv file containing the gap free AP stimulation.*

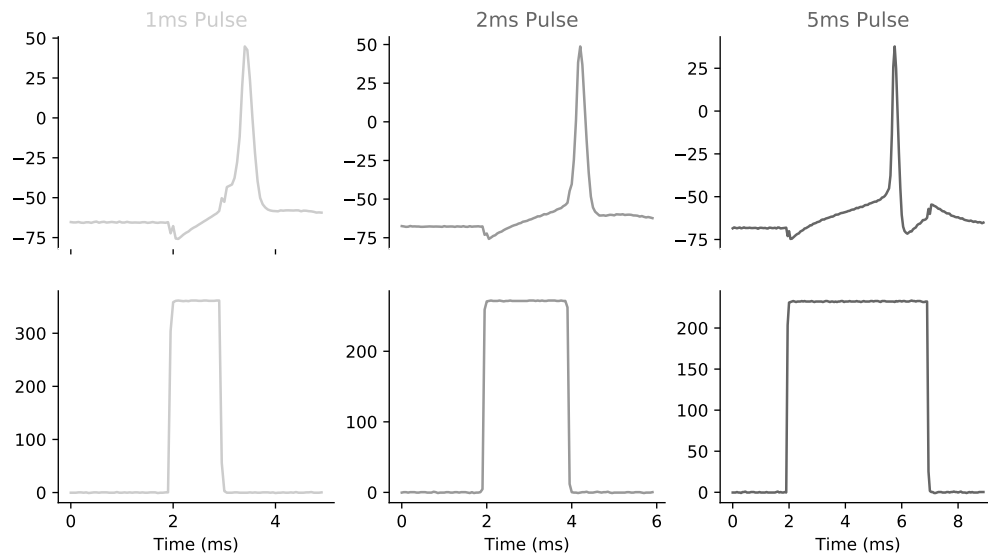
Column summary statistics:

	count	mean	std	min	25%	50%	75%	max
0	8	3.5000	2.4495	0.0000	1.7500	3.5000	5.2500	7.0000

Figures

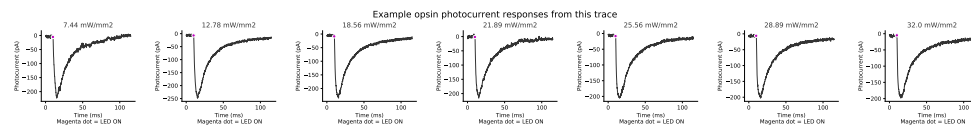
figures/figure4_1.pdf

Author comment: *Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)*



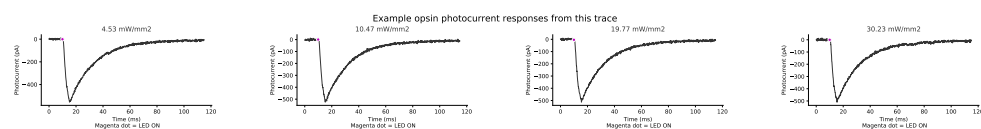
figures/figure4_2.pdf

Author comment: *Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chrimson)*



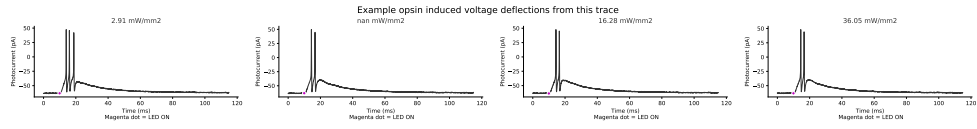
figures/figure4_3.pdf

Author comment: *Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)*



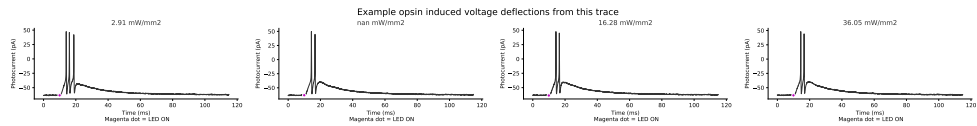
figures/figure5_1.pdf

Author comment: *Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)*



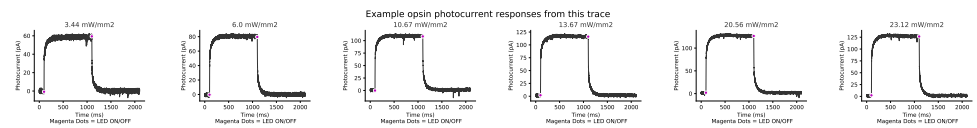
figures/figure5_1.pdf

Author comment: *Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chrimson)*



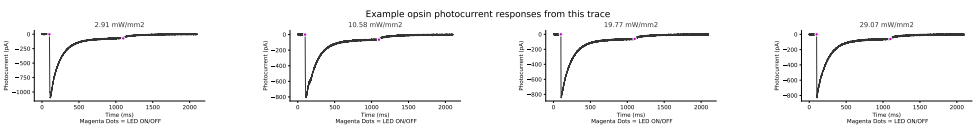
figures/figure8_1.pdf

Author comment: *Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)*



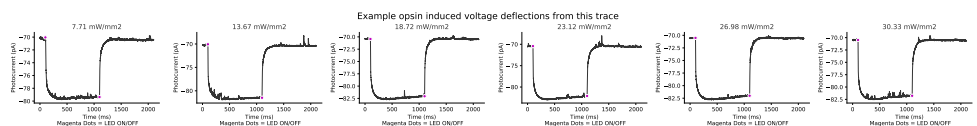
figures/figure8_2.pdf

Author comment: *Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)*



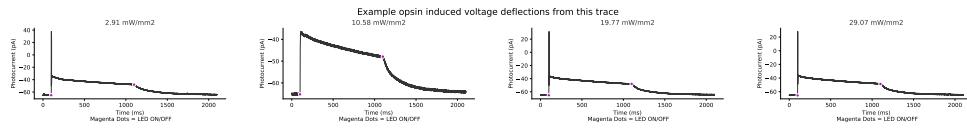
figures/figure9_1.pdf

Author comment: *Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)*



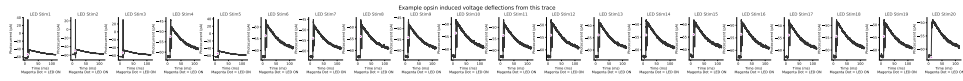
figures/figure9_2.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)*



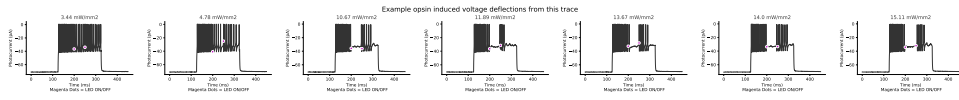
figures/figure9_3.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)*



figures/figure9_4.pdf

Author comment: *Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)*



figures/figure9_5.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)*

